(d) First, let us use (2.41) and (2.42) to calculate $\langle \psi + \chi \mid \psi + \chi \rangle$:

$$\begin{aligned} \langle \psi + \chi \mid \psi + \chi \rangle &= [(-1-3i)\langle \phi_1 \mid + 5i \langle \phi_2 \mid][(-1+3i) \mid \phi_1\rangle - 5i \mid \phi_2\rangle] \\ &= (-1-3i)(-1+3i) + (5i)(-5i) \\ &= 35. \end{aligned} \tag{2.48}$$

Since $\langle \psi \mid \psi \rangle = 58$ and $\langle \chi \mid \chi \rangle = 5$, we infer that the triangle inequality (2.36) is satisfied:

$$\sqrt{35} < \sqrt{58} + \sqrt{5} \Longrightarrow \sqrt{\langle \psi + \chi \mid \psi + \chi \rangle} < \sqrt{\langle \psi \mid \psi \rangle} + \sqrt{\langle \chi \mid \chi \rangle}. \tag{2.49}$$

Example 2.5
Consider two states $|\psi_1\rangle = 2i|\phi_1\rangle + |\phi_2\rangle - a|\phi_3\rangle + 4|\phi_4\rangle$ and $|\psi_2\rangle = 3|\phi_1\rangle - i|\phi_2\rangle + 5|\phi_3\rangle - |\phi_4\rangle$, where $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, and $|\phi_4\rangle$ are orthonormal kets, and where a is a constant. Find the value of a so that $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal.

Solution
For the states $|\psi_1\rangle$ and $|\psi_2\rangle$ to be orthogonal, the scalar product $\langle \psi_2 \mid \psi_1 \rangle$ must be zero. Using the relation $\langle \psi_2 \mid = 3\langle \phi_1| + i\langle \phi_2| + 5\langle \phi_3| - \langle \phi_4|$, we can easily find the scalar product

$$\begin{aligned} \langle \psi_2 \mid \psi_1 \rangle &= (3\langle \phi_1| + i\langle \phi_2| + 5\langle \phi_3| - \langle \phi_4|)\,(2i|\phi_1\rangle + |\phi_2\rangle - a|\phi_3\rangle + 4|\phi_4\rangle) \\ &= 7i - 5a - 4. \end{aligned} \tag{2.50}$$

Since $\langle \psi_2 \mid \psi_1 \rangle = 7i - 5a - 4 = 0$, the value of a is $a = (7i-4)/5$.

2.4 Operators

2.4.1 General Definitions

Definition of an operator: An operator[1] $\hat{A}$ is a *mathematical rule* that when applied to a ket $\mid \psi\rangle$ transforms it into another ket $\mid \psi'\rangle$ of the same space and when it acts on a bra $\langle \phi \mid$ transforms it into another bra $\langle \phi' \mid$:

$$\hat{A} \mid \psi\rangle = \mid \psi'\rangle, \qquad \langle \phi \mid \hat{A} = \langle \phi' \mid . \tag{2.51}$$

A similar definition applies to wave functions:

$$\hat{A}\psi(\vec{r}) = \psi'(\vec{r}), \qquad \phi(\vec{r})\hat{A} = \phi'(\vec{r}). \tag{2.52}$$

Examples of operators
Here are some of the operators that we will use in this text:

- Unity operator: it leaves any ket unchanged, $\hat{I} \mid \psi\rangle = \mid \psi\rangle$.
- The gradient operator: $\vec{\nabla}\psi(\vec{r}) = (\partial\psi(\vec{r})/\partial x)\vec{i} + (\partial\psi(\vec{r})/\partial y)\vec{j} + (\partial\psi(\vec{r})/\partial z)\vec{k}$.

[1] The hat on $\hat{A}$ will be used throughout this text to distinguish an operator $\hat{A}$ from a complex number or a matrix A.

- The linear momentum operator: $\vec{P}\psi(\vec{r}) = -i\hbar\vec{\nabla}\psi(\vec{r})$.
- The Laplacian operator: $\nabla^2\psi(\vec{r}) = \partial^2\psi(\vec{r})/\partial x^2 + \partial^2\psi(\vec{r})/\partial y^2 + \partial^2\psi(\vec{r})/\partial z^2$.
- The parity operator: $\hat{\mathcal{P}}\psi(\vec{r}) = \psi(-\vec{r})$.

Products of operators
The product of two operators is generally not commutative:

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}. \tag{2.53}$$

The product of operators is, however, associative:

$$\hat{A}\hat{B}\hat{C} = \hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C}. \tag{2.54}$$

We may also write $\hat{A}^n\hat{A}^m = \hat{A}^{n+m}$. When the product $\hat{A}\hat{B}$ operates on a ket $\mid \psi\rangle$ (the order of application is important), the operator $\hat{B}$ acts first on $\mid \psi\rangle$ and then $\hat{A}$ acts on the new ket $(B \mid \psi\rangle)$:

$$\hat{A}\hat{B} \mid \psi\rangle = \hat{A}(\hat{B} \mid \psi\rangle). \tag{2.55}$$

Similarly, when $\hat{A}\hat{B}\hat{C}\hat{D}$ operates on a ket $\mid \psi\rangle$, $\hat{D}$ acts first, then $\hat{C}$, then $\hat{B}$, and then $\hat{A}$.

When an operator $\hat{A}$ is sandwiched between a bra $\langle\phi \mid$ and a ket $\mid \psi\rangle$, it yields in general a complex number: $\langle\phi \mid \hat{A} \mid \psi\rangle$ = complex number. The quantity $\langle\phi \mid \hat{A} \mid \psi\rangle$ can also be a purely real or a purely imaginary number. **Note:** In evaluating $\langle\phi \mid \hat{A} \mid \psi\rangle$ it does not matter if one first applies $\hat{A}$ to the ket and then takes the bra-ket or one first applies $\hat{A}$ to the bra and then takes the bra-ket; that is $(\langle\phi \mid \hat{A}) \mid \psi\rangle = \langle\phi \mid (\hat{A} \mid \psi\rangle)$.

Linear operators
An operator $\hat{A}$ is said to be *linear* if it obeys the distributive law and, like all operators, it commutes with constants. That is, an operator $\hat{A}$ is linear if, for any vectors $\mid \psi_1\rangle$ and $\mid \psi_2\rangle$ and any complex numbers a_1 and a_2, we have

$$\hat{A}\,(a_1 \mid \psi_1\rangle + a_2 \mid \psi_2\rangle) = a_1\hat{A} \mid \psi_1\rangle + a_2\hat{A} \mid \psi_2\rangle, \tag{2.56}$$

and

$$(\langle\psi_1 \mid a_1 + \langle\psi_2 \mid a_2)\hat{A} = a_1\langle\psi_1 \mid \hat{A} + a_2\langle\psi_2 \mid \hat{A}. \tag{2.57}$$

Remarks

- The *expectation* or *mean* value $\langle\hat{A}\rangle$ of an operator $\hat{A}$ with respect to a state $\mid \psi\rangle$ is defined by

$$\langle\hat{A}\rangle = \frac{\langle\psi \mid \hat{A} \mid \psi\rangle}{\langle\psi \mid \psi\rangle}. \tag{2.58}$$

- The quantity $\mid \phi\rangle\langle\psi \mid$ (i.e., the product of a ket with a bra) is a linear operator in Dirac's notation. To see this, when $\mid \phi\rangle\langle\psi \mid$ is applied to a ket $\mid \psi'\rangle$, we obtain another ket:

$$\mid \phi\rangle\langle\psi \mid \psi'\rangle = \langle\psi \mid \psi'\rangle \mid \phi\rangle, \tag{2.59}$$

  since $\langle\psi \mid \psi'\rangle$ is a complex number.

- Products of the type $\mid \psi\rangle\hat{A}$ and $\hat{A}\langle\psi \mid$ (i.e., when an operator stands on the right of a ket or on the left of a bra) are forbidden. They are not operators, or kets, or bras; they have no mathematical or physical meanings (see equation (2.219) for an illustration).

2.4.2 Hermitian Adjoint

The Hermitian adjoint or conjugate[2], $\alpha^\dagger$, of a complex number α is the complex conjugate of this number: $\alpha^\dagger = \alpha^*$. The Hermitian adjoint, or simply the adjoint, $\hat{A}^\dagger$, of an operator $\hat{A}$ is defined by this relation:

$$\langle \psi \mid \hat{A}^\dagger \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*. \tag{2.60}$$

Properties of the Hermitian conjugate rule
To obtain the Hermitian adjoint of any expression, we must cyclically reverse the order of the factors and make three replacements:

- Replace constants by their complex conjugates: $\alpha^\dagger = \alpha^*$.
- Replace kets (bras) by the corresponding bras (kets): $(\mid \psi \rangle)^\dagger = \langle \psi \mid$ and $(\langle \psi \mid)^\dagger = \mid \psi \rangle$.
- Replace operators by their adjoints.

Following these rules, we can write

$$\begin{aligned}
(\hat{A}^\dagger)^\dagger &= \hat{A}, && (2.61)\\
(a\hat{A})^\dagger &= a^* \hat{A}^\dagger, && (2.62)\\
(\hat{A}^n)^\dagger &= (\hat{A}^\dagger)^n, && (2.63)\\
(\hat{A} + \hat{B} + \hat{C} + \hat{D})^\dagger &= \hat{A}^\dagger + \hat{B}^\dagger + \hat{C}^\dagger + \hat{D}^\dagger, && (2.64)\\
(\hat{A}\hat{B}\hat{C}\hat{D})^\dagger &= \hat{D}^\dagger \hat{C}^\dagger \hat{B}^\dagger \hat{A}^\dagger, && (2.65)\\
(\hat{A}\hat{B}\hat{C}\hat{D} \mid \psi\rangle)^\dagger &= \langle \psi \mid D^\dagger C^\dagger B^\dagger A^\dagger. && (2.66)
\end{aligned}$$

The Hermitian adjoint of the operator $\mid \psi\rangle\langle\phi \mid$ is given by

$$(\mid \psi\rangle\langle\phi \mid)^\dagger = \mid \phi\rangle\langle\psi \mid . \tag{2.67}$$

Operators act inside kets and bras, respectively, as follows:

$$\mid \alpha \hat{A}\psi\rangle = \alpha \hat{A} \mid \psi\rangle, \qquad \langle \alpha \hat{A}\psi \mid = \alpha^* \langle \psi \mid \hat{A}^\dagger. \tag{2.68}$$

Note also that $\langle \alpha \hat{A}^\dagger \psi \mid = \alpha^* \langle \psi \mid (\hat{A}^\dagger)^\dagger = \alpha^* \langle \psi \mid \hat{A}$. Hence, we can also write:

$$\langle \psi \mid \hat{A} \mid \phi \rangle = \langle \hat{A}^\dagger \psi \mid \phi \rangle = \langle \psi \mid \hat{A}\phi \rangle. \tag{2.69}$$

Hermitian and skew-Hermitian operators
An operator $\hat{A}$ is said to be *Hermitian* if it is equal to its adjoint $\hat{A}^\dagger$:

$$\boxed{\hat{A} = \hat{A}^\dagger \quad \text{or} \quad \langle \psi \mid \hat{A} \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*.} \tag{2.70}$$

[2]The terms "adjoint" and "conjugate" are used indiscriminately.

On the other hand, an operator $\hat{B}$ is said to be *skew-Hermitian* or *anti-Hermitian* if

$$\boxed{\hat{B}^\dagger = -\hat{B} \qquad \text{or} \quad \langle\psi \mid \hat{B} \mid \phi\rangle = -\langle\phi \mid \hat{B} \mid \psi\rangle^*.} \tag{2.71}$$

Remark

The Hermitian adjoint of an operator is not, in general, equal to its complex conjugate: $\hat{A}^\dagger \neq \hat{A}^*$.

Example 2.6

(a) Discuss the hermiticity of the operators $(\hat{A} + \hat{A}^\dagger)$, $i(\hat{A} + \hat{A}^\dagger)$, and $i(\hat{A} - \hat{A}^\dagger)$.

(b) Find the Hermitian adjoint of $f(\hat{A}) = (1 + i\hat{A} + 3\hat{A}^2)(1 - 2i\hat{A} - 9\hat{A}^2)/(5 + 7\hat{A})$.

(c) Show that the expectation value of a Hermitian operator is real and that of an anti-Hermitian operator is imaginary.

Solution

(a) The operator $\hat{B} = \hat{A} + \hat{A}^\dagger$ is Hermitian regardless of whether or not $\hat{A}$ is Hermitian, since

$$\hat{B}^\dagger = (\hat{A} + \hat{A}^\dagger)^\dagger = \hat{A}^\dagger + \hat{A} = \hat{B}. \tag{2.72}$$

Similarly, the operator $i(\hat{A} - \hat{A}^\dagger)$ is also Hermitian; but $i(\hat{A} + \hat{A}^\dagger)$ is anti-Hermitian, since $[i(\hat{A} + \hat{A}^\dagger)]^\dagger = -i(\hat{A} + \hat{A}^\dagger)$.

(b) Since the Hermitian adjoint of an operator function $f(\hat{A})$ is given by $f^\dagger(\hat{A}) = f^*(\hat{A}^\dagger)$, we can write

$$\left(\frac{(1 + i\hat{A} + 3\hat{A}^2)(1 - 2i\hat{A} - 9\hat{A}^2)}{5 + 7\hat{A}}\right)^\dagger = \frac{(1 + 2i\hat{A}^\dagger - 9\hat{A}^{\dagger 2})(1 - i\hat{A}^\dagger + 3\hat{A}^{\dagger 2})}{5 + 7\hat{A}^\dagger}. \tag{2.73}$$

(c) From (2.70) we immediately infer that the expectation value of a Hermitian operator is real, for it satisfies the following property:

$$\langle\psi \mid \hat{A} \mid \psi\rangle = \langle\psi \mid \hat{A} \mid \psi\rangle^*; \tag{2.74}$$

that is, if $\hat{A}^\dagger = \hat{A}$ then $\langle\psi \mid \hat{A} \mid \psi\rangle$ is real. Similarly, for an anti-Hermitian operator, $\hat{B}^\dagger = -\hat{B}$, we have

$$\langle\psi \mid \hat{B} \mid \psi\rangle = -\langle\psi \mid \hat{B} \mid \psi\rangle^*, \tag{2.75}$$

which means that $\langle\psi \mid \hat{B} \mid \psi\rangle$ is a purely imaginary number.

2.4.3 Projection Operators

An operator $\hat{P}$ is said to be a *projection operator* if it is Hermitian and equal to its own square:

$$\hat{P}^\dagger = \hat{P}, \qquad \hat{P}^2 = \hat{P}. \tag{2.76}$$

The unit operator $\hat{I}$ is a simple example of a projection operator, since $\hat{I}^\dagger = \hat{I}, \quad \hat{I}^2 = \hat{I}$.